# Java Variable

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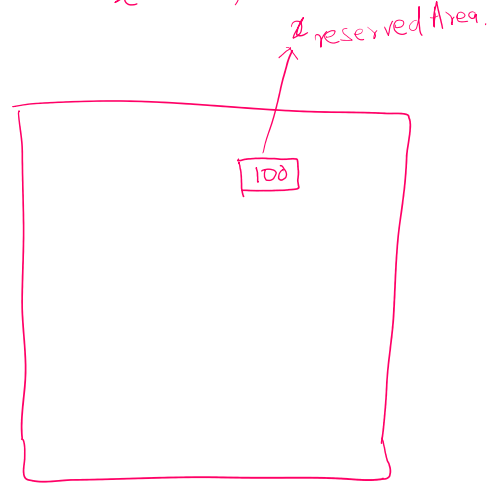
Unknown quantities are represented by the variables.
In programming, a variable is a container which holds the values, while the Java program is executed.
A variable is assigned with data type.

Variable is a name of memory location.
There are 3 types of variable in Java:
⇒ local variable.
⇒ instance variable.
⇒ static variable.

x = 10;
x = 100;




A variable is the name of a reserved area allocated in memory.
It is a combination of "vary + able" which means its value can be changed.

```
public class Q1
{
    public static void main(String args[]){
        int x = 100;
        System.out.print("\fx = " + x);
    }
}
```

data type — what kind of data we can store in the variable.
variable
it will give you fresh terminal window
To display information on the terminal window.

```
System.out.print("\fx = " + x);
```

command
value will come
join
x =

## Data types in Java

Data types specify the different sizes and values that can be stored in the variable.
There are two types of data types in Java:
1. Primitive data types: The primitive data types include boolean, byte, short, int,

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1. Primitive data types: The primitive data types include boolean, byte, short, int, long, float, double, and char.

2. Non-primitive data types or user-defined types: The non-primitive data-types are Classes, Interfaces and Arrays, etc.

## Java Primitive Data Types

In Java language, primitive data types are the building blocks of data manipulation. These are the most basic data types available in Java language.

There are 8 types of primitive data types:

⇒ boolean → 1 bit memory value (true/false) (1 / 0)

4 Integer types:
- byte → 1 Byte memory value (-128 to 127)
- short → 2 Bytes memory value (-32768 to 32767)
- int → 4 Bytes memory value (-2147483648 to 2147483647)
- long → 8 Bytes memory value (-9,223,372,036,854,775,808 to 9,223,372,036,854,775,807)

2 Floating points:
- float → 4 Bytes memory (6 decimal places)
- double → 8 Bytes memory (15 decimal places)

Character { char → 2 bytes memory
↑ it supports UNICODE (multilingual character values).

```
public class Q1
{
   public static void main(String args[]){
      int x = 100;
      System.out.print("\fx = " + x);
      boolean b = true;
      System.out.print("\nb = " + b);
      char c = 'A';
      System.out.print("\nc = " + c);
   }
}
```

fresh terminal window or new

new line. (move the printing cursor to next line)

c = 'क';

"\u ----"

```
System.out.print("\nm = " + m + "\nn = " + n + "\nr = " + r);
```

← we can write single print statement to output in separate lines by using "\n".

```
System.out.print("\nm = " + m + "\nn = " + n + "\nr = " + r);
```

We can write single print statement to output in separate lines. by using "\n".

Escape sequence.

This is the combination of two characters. first one is '\' (back slash) and next one is small alphabet.

This sequence is always used inside single quotes or double quotes.

These are called non-graphical characters.

Therefore these are commands, which calls routines to perform specific activities.

- "\f"
- "\n"
- "\u----"
- "\t"
- "\r"